

provide the Day 3 contents

Absolutely. Since **Day 1 = Networking fundamentals/OSI** and **Day 2 = complete packet-flow + TCP/IP + MAC/IP + subnetting + gateway**, Day 3 should move into **TCP/IP fundamentals** and **ARP/Switching** in depth.

July

17

Day 3 — ARP, Ethernet, Switching & Routing Fundamentals

Duration: 2.5 hours

Day 3 Goal

By the end of Day 3, you should be able to explain:

"PC-A wants to communicate with PC-B. How does it find the MAC address, how does the switch forward the frame, and how does the router forward packets when PC-B is on another network?"



Day 3 — 2.5 Hour Schedule

Time	Topic	Goal
0:00–0:20	Quick Day 2 revision	IP, subnet, gateway
0:20–0:55	ARP in detail	IP → MAC resolution
0:55–1:25	Ethernet Frame	MAC + frame structure
1:25–1:55	Switching	MAC table + forwarding
1:55–2:20	Router forwarding	Routing + next hop
2:20–2:30	Interview questions	Test yourself

Part 1 — Quick Revision

Before starting Day 3, remember this:

Application

↓



TCP
↓
IP
↓
Ethernet
↓
NIC
↓
Network

And:

IP = Where?
MAC = Which local device?
Port = Which application?

For example:

PC-A

IP = 192.168.1.10
MAC = AA:AA:AA:AA:AA:AA
Port = 50000

PC-B:

IP = 192.168.1.20
MAC = BB:BB:BB:BB:BB:BB
Port = 8080

Part 2 — ARP in Detail

2.1 What problem does ARP solve?

Suppose PC-A knows:

Destination IP = 192.168.1.20

But Ethernet requires:

Destination MAC = ??????

So PC-A needs:

192.168.1.20
↓

?

↓

BB:BB:BB:BB:BB:BB

ARP solves this:

ARP maps an IPv4 address to a MAC address on the local network.

2.2 ARP Request

PC-A broadcasts:

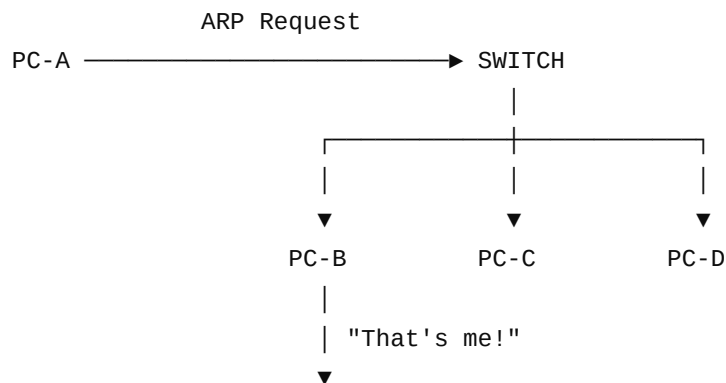
Who has 192.168.1.20?
Tell 192.168.1.10

Ethernet destination:

FF:FF:FF:FF:FF:FF

That's the Ethernet broadcast MAC.

Visual:

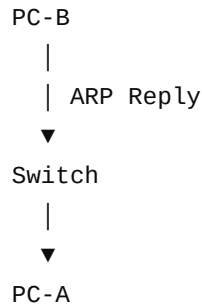


2.3 ARP Reply

PC-B responds:

192.168.1.20 is at
BB:BB:BB:BB:BB:BB

This is normally sent as a unicast Ethernet frame back to PC-A.



Now PC-A knows:

192.168.1.20
↓
BB:BB:BB:BB:BB:BB

2.4 ARP Cache

Linux doesn't want to perform ARP every time.

It stores the mapping temporarily.

Run:

```
ip neigh
```

You may see something like:

```
192.168.1.1 dev eth0 lladdr 11:22:33:44:55:66 REACHABLE
192.168.1.20 dev eth0 lladdr aa:bb:cc:dd:ee:ff REACHABLE
```

Conceptually:

IP Address	MAC Address
192.168.1.1	11:22:33:44:55:66
192.168.1.20	AA:BB:CC:DD:EE:FF

Part 3 — ARP for Different Network

This is one of the **most important interview concepts**.

Suppose:

PC-A = 192.168.1.10/24

PC-B = 10.0.0.20/24

Gateway = 192.168.1.1

PC-A determines:

192.168.1.10 → Network 192.168.1.0

10.0.0.20 → Network 10.0.0.0

Different network.

Therefore:

PC-A

|

| Destination = 10.0.0.20

▼

Routing decision

|

▼

Gateway = 192.168.1.1

|

▼

ARP for 192.168.1.1

NOT:

✗ ARP for 10.0.0.20

Instead:

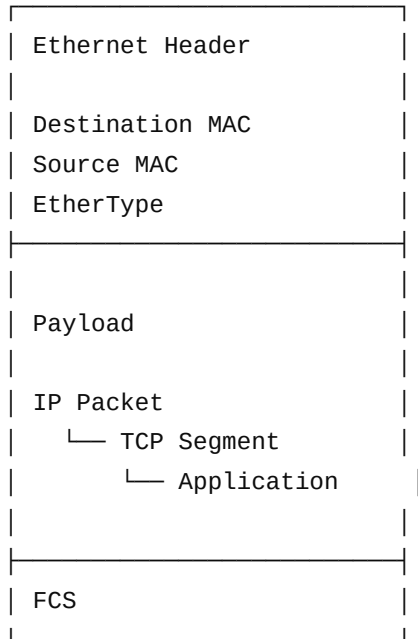
✓ ARP for 192.168.1.1

Because the Ethernet frame is being sent to the **next hop**.

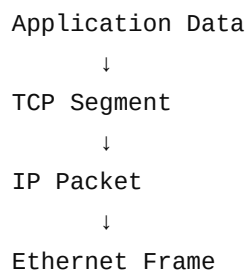
Part 4 — Ethernet Frame

Once PC-A knows the destination MAC, it creates an Ethernet frame.

Basic structure:



So:



Part 5 — Ethernet MAC Addresses

Example:

Source MAC:
AA:AA:AA:AA:AA:AA

Destination MAC:
BB:BB:BB:BB:BB:BB

For same LAN:

PC-A MAC → PC-B MAC

For different network:

PC-A MAC → Router MAC

After router:

Router MAC → PC-B MAC

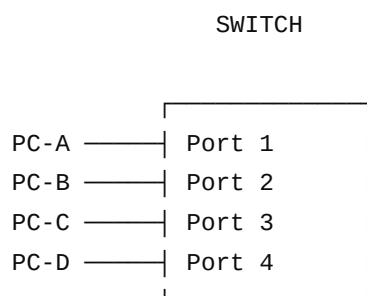
This gives you the key rule:

MAC addresses are hop-by-hop.

Part 6 — Switch

Now we need to understand exactly how the switch works.

Suppose:



The switch maintains a MAC address table:

MAC Address	Port
AA:AA:AA:AA:AA:AA	1
BB:BB:BB:BB:BB:BB	2
CC:CC:CC:CC:CC:CC	3
DD:DD:DD:DD:DD:DD	4

Part 7 — How Does a Switch Learn?

This is very important.

Suppose PC-A sends:

Source MAC = AA
Destination MAC = BB

The switch receives the frame on Port 1.

It learns:

AA → Port 1

Then checks:

Destination = BB

If the table says:

BB → Port 2

it forwards the frame only to Port 2.

```
PC-A
|
| Port 1
▼
SWITCH
|
| Port 2
▼
PC-B
```

Part 8 — What if Destination MAC is Unknown?

Suppose:

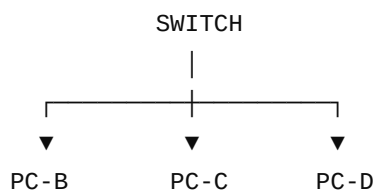
Destination MAC = XX

and switch doesn't know:

XX → ?

The switch performs **unknown unicast flooding** within the relevant VLAN.

Conceptually:



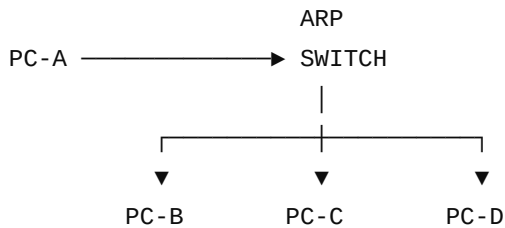
The correct destination responds, and the switch can learn its MAC from the source MAC of future frames.

Part 9 — Broadcast

ARP Request is broadcast:

FF:FF:FF:FF:FF:FF

The switch forwards it out the other ports in the same broadcast domain/VLAN.



Routers normally **do not forward Layer-2 broadcasts** between interfaces.

This is why ARP is local to a Layer-2 network.

Part 10 — Router

Now connect the switch to the router.



The router looks at:

Destination IP

For example:

10.0.0.20

Then:

Routing Table

↓

10.0.0.0/24

↓

eth1

Router forwards the packet through eth1 .

Part 11 — Router Changes Layer 2

Incoming:

Ethernet

Src MAC = PC-A

Dst MAC = Router

Router removes/processes the incoming Layer-2 framing.

The IP packet remains the Layer-3 forwarding object.

Then the router creates a new Layer-2 frame:

Ethernet

Src MAC = Router

Dst MAC = PC-B

So:

IP
PC-A → PC-B

MAC
PC-A → Router → PC-B

Part 12 — TTL

Another Day-3 interview point.

Suppose:

TTL = 64

When a router forwards the packet:

64
↓
63
↓
62
↓
...

Every Layer-3 router hop decrements TTL.

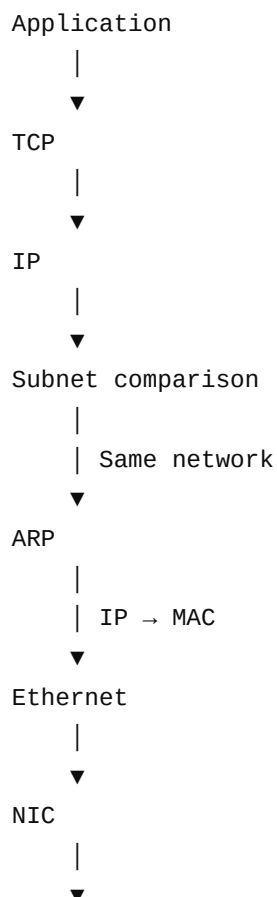
If TTL reaches zero, the packet is discarded and the router normally sends an ICMP Time Exceeded message.

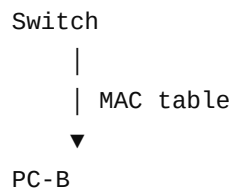
This prevents packets from circulating forever because of routing loops.

Part 13 — Complete Flow

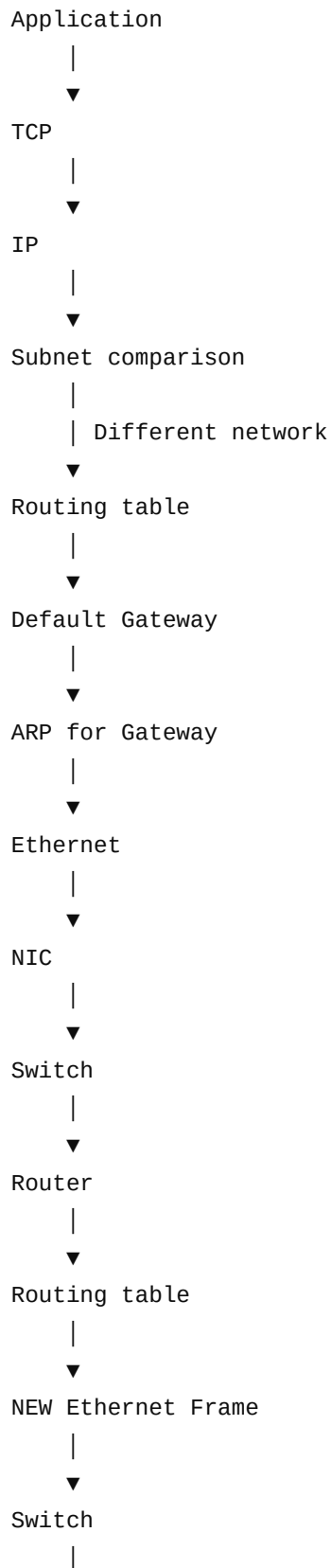
Now combine Day 2 + Day 3.

Same LAN





Different network





Part 14 — Important Commands for Hands-On

Since you're learning Linux networking, run these on Ubuntu.

1. IP address

```
ip addr
```

Look for:

```
inet 192.168.1.10/24
```

2. Routing table

```
ip route
```

Look for:

```
default via 192.168.1.1  
192.168.1.0/24 dev eth0
```

3. ARP/neighbor table

```
ip neigh
```

4. Test ARP

```
ping 192.168.1.1
```

Then:

```
ip neigh
```

You should see a neighbor entry if the gateway is reachable and ARP has been resolved.

5. Observe packets

Install/use:

```
sudo tcpdump -i any
```

Then in another terminal:

```
ping 192.168.1.1
```

For ARP specifically:

```
sudo tcpdump -i any arp
```

For TCP:

```
sudo tcpdump -i any tcp
```

This is where your theoretical model becomes much easier to remember.

Part 15 — Day 3 Interview Questions

You should be able to answer these without looking at notes:

Beginner

1. What is ARP?
2. Why is ARP required?
3. What is an ARP Request?

4. What is an ARP Reply?
5. What is an ARP cache?
6. What is a MAC address?
7. What is an Ethernet frame?
8. What does a switch use to forward a frame?
9. What is a MAC address table?
10. How does a switch learn MAC addresses?

Intermediate

11. What happens when the switch doesn't know the destination MAC?
12. Why is an ARP request broadcast?
13. Does a router forward ARP broadcasts?
14. When communicating with a remote network, whose MAC address does the host need?
15. Why doesn't PC-A ARP for a remote PC?
16. What is a default gateway?
17. How does a router select the outgoing interface?
18. What happens to the Ethernet frame at a router?
19. Why do MAC addresses change at every router hop?
20. What happens to the destination IP when a router forwards a packet?

Linux interview

21. How do you view the IP address?

```
ip addr
```

22. How do you view the routing table?

```
ip route
```

23. How do you view the ARP/neighbor table?

```
ip neigh
```

24. How can you capture ARP packets?

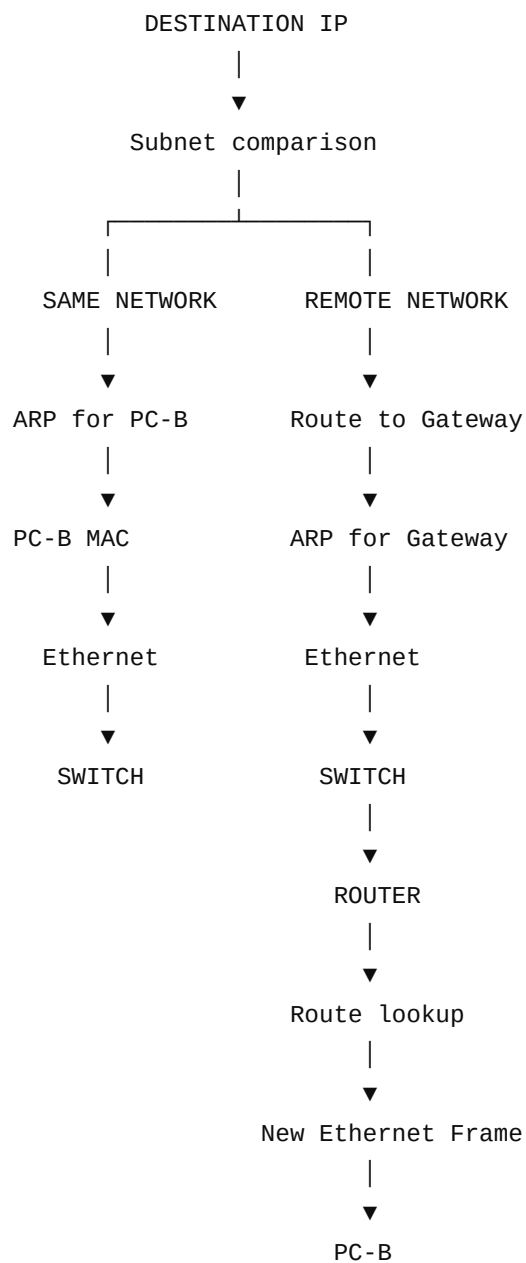
```
sudo tcpdump -i any arp
```

25. How can you capture TCP packets?

```
sudo tcpdump -i any tcp
```

★ Day 3 Must-Know Summary

Memorize this chain:



Five rules for the interview

1. IP decides the network path.
2. ARP resolves an IPv4 next-hop address to a MAC address.
3. Switches forward Ethernet frames using MAC addresses.
4. Routers forward IP packets using routing tables.
5. MAC addresses are hop-by-hop; IP addresses are normally end-to-end.

Once these are solid, **Day 4 should move into TCP deeply: TCP connection states, sequence numbers, ACKs, retransmission, flow control, congestion control, and the detailed `connect()` → 3-way handshake → `send()` → `recv()` flow.**